

XII – Computer Science – Cycle Test-1

Marks: 20

1. What is a function in Python? (1 mark)
2. What are the three types of functions? Give examples for each. (3 marks)
3. Consider the following code find and write the following parts: (4 marks)
 - a. Function definition
 - b. Function call
 - c. Formal parameter
 - d. Actual argument
 - e. Default parameter
 - g. Positional argument
 - h. Non-default parameter
 - i. Keyword argument

```
def fun (a, b=10):
    add=a+b
    print (add)
fun (50, b=20)
```

4. What is the output of the following codes: (6X2=12 marks)

i. p=5

```
def sum(q,r=2):
    global p
    p=r+q**2
    print(p, end= '#')
a=10
b=5
sum(a,b)
sum(r=5,q=1)
```

ii. def power(b,p):

```
y=b**p
return y
def calcSquare(x):
    a=power(x,2)
    return a
n=5
result=calcSquare(n)+power(3,3)
print("The Final Value is:")
print(result)
```

iii. def Change(P,Q=30):

```
P=P+Q
Q=P-Q
print(P,"#",Q)
return P
R=150
S=100
R=Change(R,S)
print(R,"#",S)
S=Change(S)
```

iv. c=10

```
def add():
    global c
    c=c+2
    print("Inside add():",c)
add()
c=15
print("In main:",c)
```

v. def prod(x,y,z):

```
print(x*y*z)
def summ(x,y,z):
    return x+y+z
a=summ(6,16,6)
b=prod(2,4,6)
print(a,b)
```

vi. def Diff(N1,N2):

```
if N1>N2:
    return N1-N2
else:
    return N2-N1
NUM= [10,23,14,54,32]
for CNT in range (4,0,-1):
    A=NUM[CNT]
    B=NUM[CNT-1]
    print(Diff(A,B),'#', end= ' ')
```